**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 135 · Network Protocols and Layers

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does a wireless access point do?

Q2. Why can a mesh survive some broken links?
A. It has only one path
B. Every device is a server
C. It has other paths
D. Every link is wireless

Q3. “Fibre carries electrical signals through copper wire.” Correct this. Explain why.

Q4. Desktops and moving tablets need network links. “Wi-Fi needs a cable from each tablet to the switch.” Correct this. Which links suit these devices?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

IP address — Identifies devices/interfaces for routing across IP networks.

MAC address — Identifies a network interface at the data-link level for local communication.

DNS — Resolves a domain name to an IP address.

Protocols and layers — Agreed rules such as HTTP(S), TCP/IP and SMTP support interoperable communication.

## New learning

IP address: Identifies devices/interfaces for routing across IP networks.

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DNS: Resolves a domain name to an IP address.

Protocols and layers: Agreed rules such as HTTP(S), TCP/IP and SMTP support interoperable communication.

## Worked example

A pupil enters a website domain name into a browser. Explain DNS and packet delivery.

1. The browser needs the server's IP address.

2. DNS provides the mapping from domain name to IP.

3. Packets can be routed to the destination across networks.

4. The browser and server use suitable protocols, e.g. HTTPS for encrypted web transfer.

# We Do · Practise with guidance

### Guided 1

Explain how a school laptop sends an email.

Worked method: Avoid confusing SMTP with IMAP/POP reception.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Complete the comparison that recognises a secure web protocol.

```
protocol = "HTTPS"
if ... then
    print("Secure web")
endif
```

write

### Question 1

State the purpose of DNS. [2 marks]

### Question 2

Distinguish IP and MAC addresses. [4 marks]

### Question 3

Name a suitable protocol for sending email. [1 mark]

# Exit Ticket · Answer before checking

DNS translates a domain name into an IP address.

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It enables wireless devices to connect to the network.

Q2. C — It has other paths

Q3. Fibre uses light. It supports fast data transfer over long distances and resists electromagnetic interference. Installation can cost more.

Q4. Ethernet uses cables. Wi-Fi uses radio, so tablets can move around, but interference can affect it. Choose links to suit the distance and data rate needed.

### I Do / We Do pages · Guided practice

Q1. Avoid confusing SMTP with IMAP/POP reception.

### You Do pages · Code First

```
protocol = "HTTPS"
if protocol == "HTTPS" then
    print("Secure web")
endif
```

Expected result: Secure web

### You Do pages · Independent questions

Q1. Resolves domain names to IP addresses.

Q2. IP used for logical addressing/routing across networks; MAC identifies a network interface for local data-link forwarding.

Q3. SMTP.

### Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## 2. Write a complete new answer in GREEN PEN. Show working if needed: